4.1, 4.3, 4.5, 4.7, 4.9, 4.13, 4.15, 4.17

4.1 Show by induction that $2^n < n!$ for all $n \geq 4$.

For the base statement, n = 4, the statement that $2^4 < (4)!$ holds true as it boils down to $16 < 24$.
An inductive proof for this could be shown below
$2^{n+1} < (n+1)!$ which can be rewritten as $2(2^n) < (n+1)!$
Since $n > 4$, then $n + 1 > 5$, which means that $n + 1 > 2$ also hold true, multiplying $n!$ results in: $2n! < (n+1)n!$
If you multiply the induction hypothesis ($2^n < n!$) by 2, you get:
$2(2^n) < 2n!$
Then combining the above inequalities yields:
$2^{n+1} < 2n! < (n+1)!$
This proves that $2^n < n!$ holds for n+1. □

4.3 Show by induction that $f_0 + ... + f_n = f_{n+2} - 1$ in the Fibonacci sequence.

First, proving the base case:
For $n = 0$, $f_n = 1$ so $f_{n+2} - 1 = 2 - 1 = 1$ This proves the base case.
Now, assuming that n is true:
$f_n = f_{n+1} - 1$
Finally, the inductive step:

so $f_{n+1} = f_{n+2} - 1$

4.5 Consider an alternative Fibonacci sequence, starting with 0, 2. Can you prove the analogue of theorem 25? Generalize the result as far as you can.

4.7 Show by induction that a finite set with n elements has exactly 2^n many subsets.

4.9 Prove that $\frac{1}{1*2} + \frac{1}{2*3} + \frac{1}{3*4} + ... + \frac{n}{n*n+1} = \frac{n}{n+1}$.

4.13 True or false: every flying polka-dotted elephant in this room is smoking a cigar. Criticize the following "proof."

Claim. $\sum_{i=1}^{\infty} \frac{1}{i} < \infty$.

Proof. **Let $S(n) = \sum_{i=1}^{n} \frac{1}{i}$, and consider the statement asserting $S(n) < \infty$. This statement is true for $n = 1$, since $\frac{1}{1} = 1 < \infty$. And if it is true for n, then it is true for $n + 1$, since the next sum $S(n+1) = \sum_{i=1}^{n+1} \frac{1}{i}$ is equal to the previous sum $\sum_{i=1}^{n} \frac{1}{i}$ plus the next term $\frac{1}{n+1}$, or in other words, $S(n+1) = S(n) + \frac{1}{n+1}$, and this is the sum of two finite numbers. So $S(1)$ is true, and $S(n)$ implies $S(n+1)$. So we have therefore proved $\sum_{i=1}^{\infty} \frac{1}{i} < \infty$ by induction.** □

4.17 Prove that the least-number principle, the common induction principle, and the strong induction principle are all equivalent. Assuming any one of them, you can prove the others.